Name & Surname **Polimi ID**

Exercise 1

A. What is the difference between a true dependence and an anti-dependence? (1)

B. What are the challenges of using a shared state in a Map function? (1)

C. Consider the following nested loop:

```
for (j = 0; j < 100; j++) {
    b[j] = f1(a[j]);
}
for (j = 99; j >= 0; j--) {
    a[j] = f2(b[j]);
}
```

Which loop can be parallelized and under which conditions (j is declared as an integer)? Is there a way to improve the code? Is the OpenMP nowait clause a possible option? Justify your answers. (2)

Name & Surname **Polimi ID**

Exercise 2

A. Please describe the parallel reduce pattern: assumptions, PRAM complexity, and example. What about the work done? Is it different from the one done by the sequential algorithm? (2)

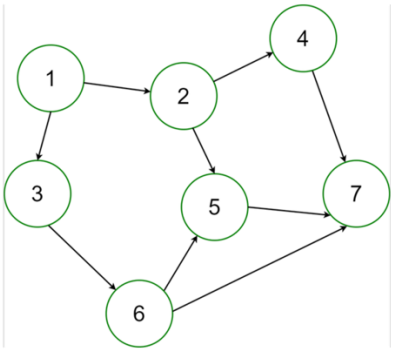
B. Please discuss how the Unpack parallel pattern works. How is it implemented (pseudo code required and PRAM-based complexity)? (2)

Name & Surname **Polimi ID**

Exercise 3

A. The input to a High-Level Synthesis tool is a Register-Transfer Level description of an accelerator. Is this statement true or false? Explain. (1)

Consider the following task graph, where all tasks contain the same amount of Work:



B. Indicate the critical path in the figure, and calculate the Parallelism. (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

C. Write a corresponding OpenMP implementation, explaining your choice of work-sharing construct. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**

Name & Surname **Polimi ID**

Exercise 4

A. **Which** software components do I get from an MPI implementation? (1)

B. The following is the `MPI_Recv` signature:

```
int MPI_Recv(void *buf, int count, MPI_Datatype datatype,
             int source, int tag, MPI_Comm comm, MPI_Status *status)
```

What value should I provide to the `count` parameter? (1)

C. Make an **example** where MPI_Comm_split is useful. Do **not** describe what the function does. Describe a scenario where it makes sense to use it, how you use it, and the reason you use it. (2)

Name & Surname **Polimi ID**